Usability Evaluation: User-based testing

Travel.State.Gov, U.S. Passport Application System

Shanikka Richardson

Study Overview

The tests for this usability study took place over the course of three (3) days (November 4th, November 6th and November 7th). Three of the user tests were completed on the same day with the following two taking place over the course of two (2) additional days. Each participant either tested alone in a remote setting or accompanied by me as the tester beside them. Each test was recorded via Google Meet, with the participants either sharing their screen or recording the screen as they went through each task.

There were five (5) participants total, ranging in demographic background, technical experience, and other considered special factors.

Participant 1: Female, Mid-70s, retired accountant, limited technical expertise, has traveled outside of the United States.

Participant 2: Female, Mid-60s, staff accountant, mid-range technical expertise, mother of 1, no previous travel outside of the United States.

Participant 3: Male, early 20s, college student, interacts with technology heavily, has never applied for a U.S. passport before.

Participant 3: Female, early 20s, web content coordinator, works with tech professionally, current passport holder.

Participant 5: Female, mid-20s, writer, interacts with technology heavily, current passport holder, and frequent traveler.

In the Design for Diversity framework, they stressed the importance of designing for our Excluded Source, as they will interact and/or be impacted by our product. For this reason, I chose **Participant 1** as I knew they were outside the Target Source or intended user group. Each participant was chosen based on their demographic or other special factors. **Participant 2** is a mother and would potentially need to access the site for their child. **Participants 3 & 4** were chosen because of their higher exposure and experience with technology and tech-based application systems. **Participant 5** was chosen due to their frequent travel and known familiarity with the site.

Two of my participants (**1 and 2**) are outside the typical user group mentioned in our first assignment. This was done intentionally to consider the D4D framework.

Task Descriptions

For the usability tests, five tasks were chosen and kept consistent across each user.

Task 1: Find the required documents to submit with your passport application.
Task 2: Locate the nearest passport acceptance agency.
Task 3: Find the current passport processing time.
Task 4: Fill out the online application form.
Task 5: Determine the next steps for submitting your passport application.

Each of these tasks is vital to the overall passport application process. Additionally, these tasks include a mix of information-gathering and actual task completion. This ensures that I can test the information architecture, language, and content relevancy of the site and the technical usability of the site tools.

For the pilot test, I recruited a user who was very familiar with the passport application process and well within our target user range. My **pilot study participant** is female, in her early 30s, a frequent international traveler, and incredibly tech savvy. In the pilot, my test user went through each task to completion. With their feedback, I could clarify task wording and narrow down which tasks were necessary and common for site users. Most importantly, I got baseline task completion metrics for user comparisons in my analysis. This helped me determine how long each task *should* take the ideal user.

Usability Test Results

Before sharing the results of the usability study, I want to clarify the terms used going forward. *User Task Completion* is the user's recognized ability to complete a task. This does not mean the task is correct or successful, just that the user feels they have completed what is asked of them. *User Task Success is* the measure of whether the task completed was done so successfully. This is very important to note, as users can **believe** that they successfully completed a task but in reality, did not.

TABLE 1: USER TASK SUCCESS

Task	P01	P02	P03	P04	P05	*% of users tested:*
1	-	-	**X**	**X**	-	**40%**
2	**X**	**X**	**X**	-	**X**	**80%**
3	-	-	**X**	**X**	-	**40%**
4	-	**X**	**X**	**X**	**X**	**80%**
5	-	**X**	**X**	**X**	**X**	**80%**
% of tasks:	**20%**	**60%**	**100%**	**80%**	**60%**	

X: task was completed successfully
- : task was not completed successfully

In Table 1, we can see which tasks each participant completed successfully as well as a percentage of how often a task was completed successfully overall. **P03** was the only participant to complete each task, while **Tasks 2**, **4,** and **5** were the most often completed. This is helpful in identifying which tasks can be more difficult for a user to complete.

From this table, we can see that **Task 1** (finding required documents) and **Task 3** (finding the current processing time) are more challenging for users to complete successfully.

Now, let's address the participants' results. **P01**, our Excluded Source, could only complete one out of five tasks successfully. There is a 40% difference between their result and the next two lowest participants. This could conclude that the site is inaccessible for this particular user or for similar users. It is also important to note that the task P01 was able to complete successfully was a common task for each participant.

In Table 2, we cover the participants' belief of task completion and to what degree they struggled with each task. In this way, we can see how confident each participant was in completing each task. **P03** and **P04** were the most confident and struggled less with their choices than other participants.

TABLE 2: USER TASK COMPLETION

TASK	P01	P02	P03	P04	P05
Find required supporting documents.	1	2	3	3	2
Locate the nearest passport agency near you.	2	3	3	3	3
Find the current passport processing time.	2	2	3	3	3
Fill out the online application form.	2	3	3	3	3
Determine how you would submit your application form.	2	3	3	3	3
SUM	9	13	15	15	14

3: User can complete task with no trouble.
2: User can complete task but has some struggles.
1: User can't complete task.

With the data in **Tables 1** and **2**, we can see how many of our participants thought that they completed the task and *actually didn't*. For example, **P01** was the only participant to stop attempting Task 1 and move on to the next task, but **P02** and **P05** also did not complete the task successfully. A usability benchmark is not just the user's belief that they can complete something but whether they do so adequately.

In **Table 3**, I measured how long it took each participant to complete each task and then compared it to our Pilot Study participant's task time.

TABLE 3: User Task Time Performance

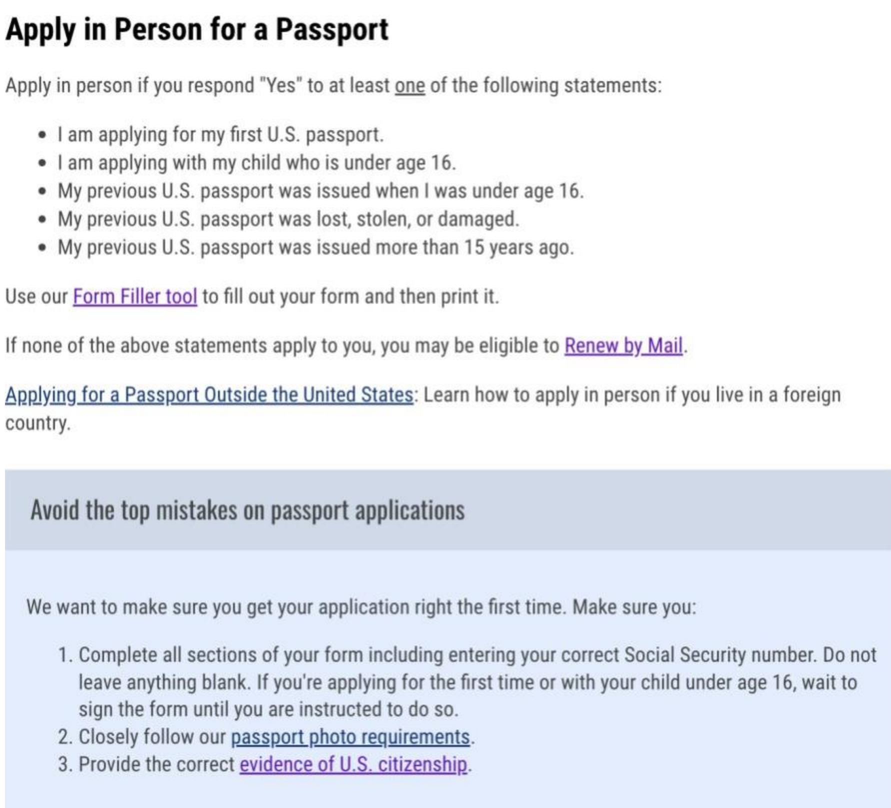
TASK	P01	P02	P03	P04	P05
Find required supporting documents.	0	2	3	2	2
Locate the nearest passport agency near you.	1	2	3	3	2
Find the current passport processing time.	1	2	3	3	2
Fill out the online application form.	1	1	2	2	2
Determine how you would submit your application form.	1	2	2	2	2
SUM	4	10	13	12	10

3: Performed the task quicker than the expected time range.
2: Performed the task within the expected time range.
1: Performed the task outside the expected time range.
0: Did not perform the task at all.

Not only was **P03** the most confident and the most successful, but he was also the closest to our expected time range for task performance. **P03** mirrors what we want our target source's experience to be on the site. While this may be the ideal performance for one user, **P01** had the most frustrating experience with this site. It took them longer to complete a task, and when they did so, they were only successful 20% of the time.

Proposed Changes Based on Study

Based on the results of my user-based testing, I would prioritize the information architecture and layout of the application information page and enhance the search bar visibility.



Apply in Person for a Passport

Apply in person if you respond "Yes" to at least one of the following statements:

- I am applying for my first U.S. passport.
- I am applying with my child who is under age 16.
- My previous U.S. passport was issued when I was under age 16.
- My previous U.S. passport was lost, stolen, or damaged.
- My previous U.S. passport was issued more than 15 years ago.

Use our Form Filler tool to fill out your form and then print it.

If none of the above statements apply to you, you may be eligible to Renew by Mail.

Applying for a Passport Outside the United States: Learn how to apply in person if you live in a foreign country.

Avoid the top mistakes on passport applications

We want to make sure you get your application right the first time. Make sure you:

1. Complete all sections of your form including entering your correct Social Security number. Do not leave anything blank. If you're applying for the first time or with your child under age 16, wait to sign the form until you are instructed to do so.
2. Closely follow our passport photo requirements.
3. Provide the correct evidence of U.S. citizenship.

Figure 1: Apply in Person information

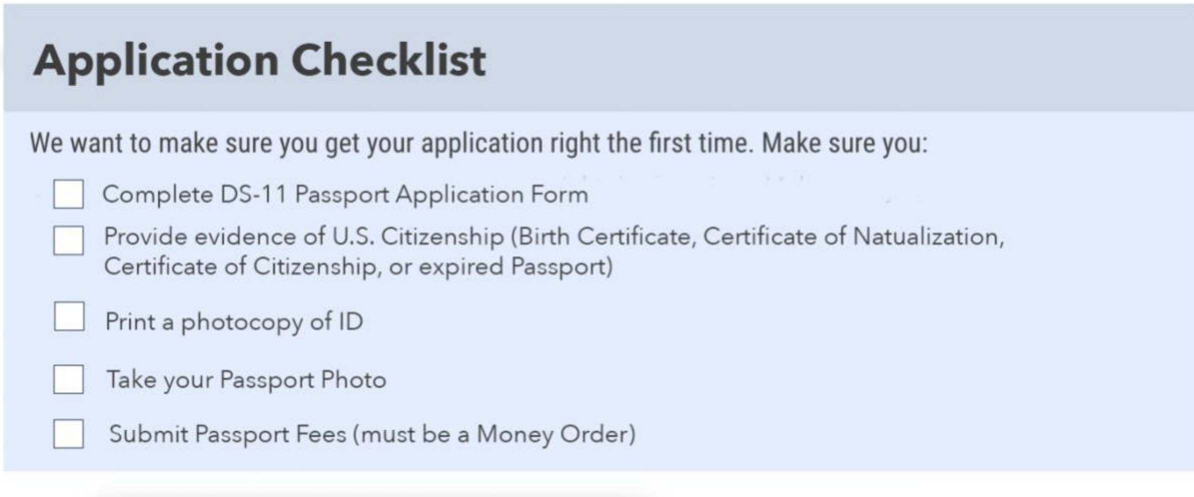


Figure 2: Application Checklist (After)

An application checklist would help the users know which documents are required as soon as they land on the page. This box is highlighted near the top of the page.

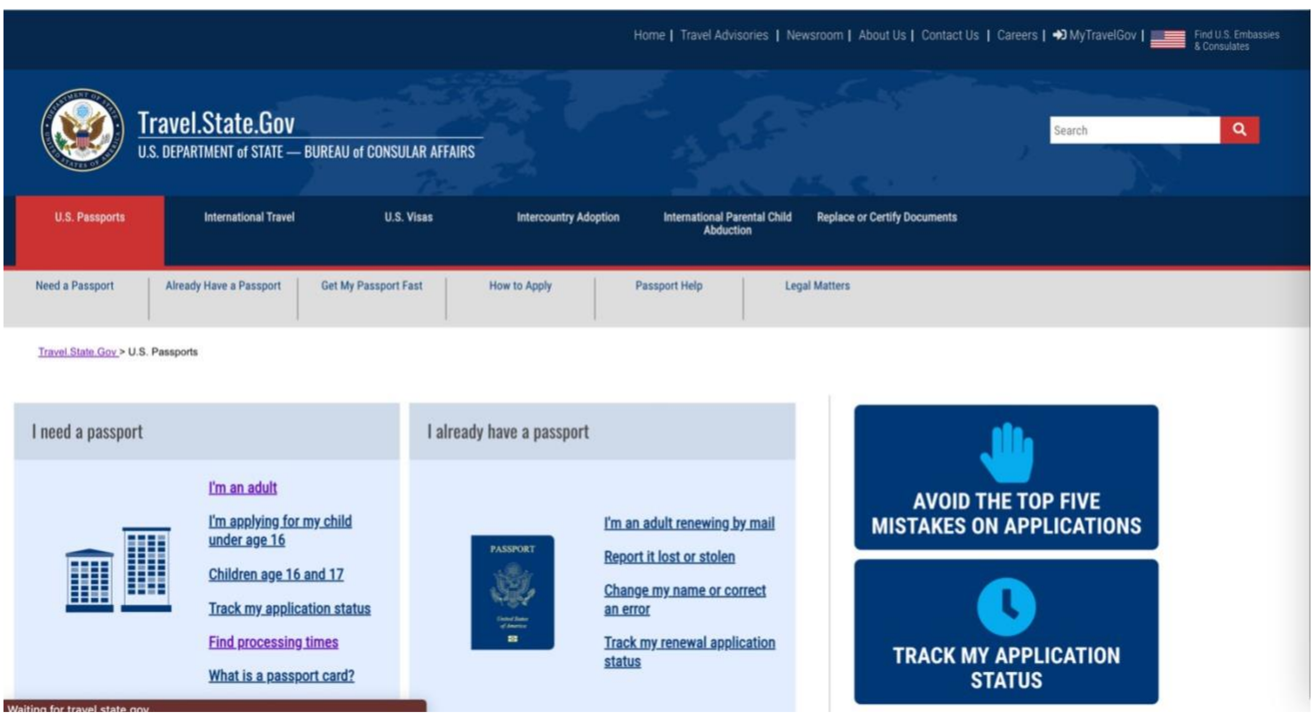


Figure 3: Search Bar Location / Increased Functionality

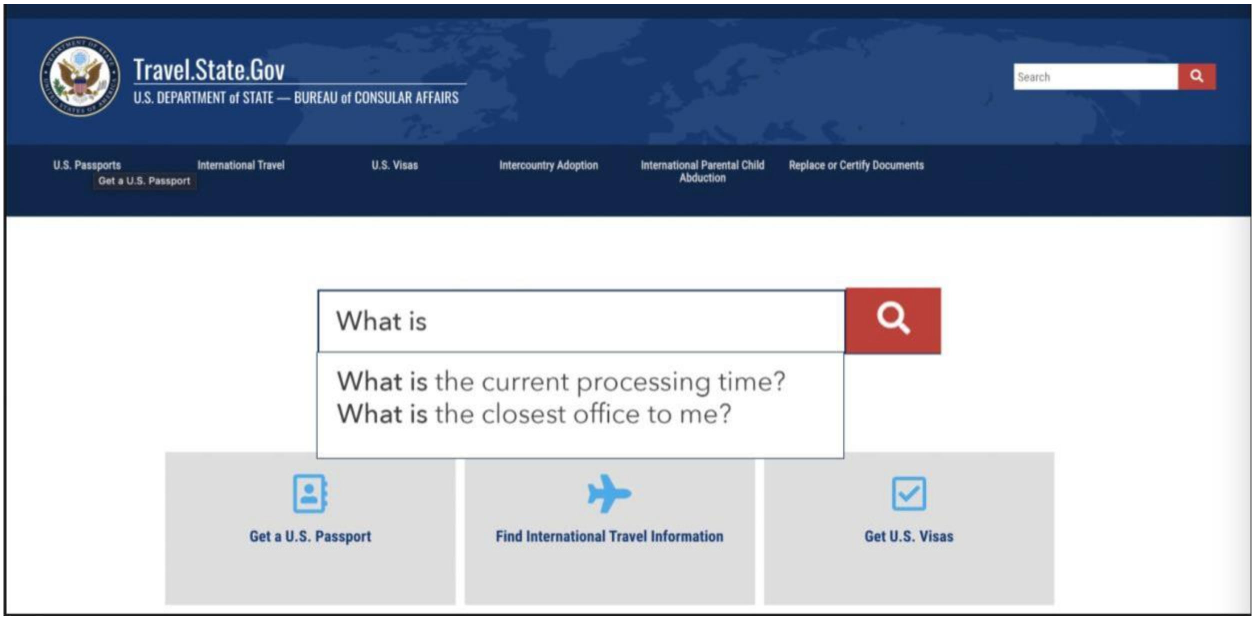


Figure 4: Search Engine Location / Functionality

A predictive and more prominent search bar will help user's find the information they are looking for more easily, even if the information is not on the page that they expect.

In my expert review, I stressed the importance of technical and visual aspects within the online application system. However, from the results of our user-based testing, most issues lie in the web content structure and information-gathering process. Users cannot locate simple application information, such as processing times and required documents. In my testing, less than half of the participants could complete either task successfully. This means that they either found incorrect information or inadequate information. In either event, their passport application would not be processed.

To improve this system, I would prioritize the following:

- **Clearer language** in the interface

  The reading level for pertinent information acted as a barrier for the users. This created uncertainty in results and even failure to task completion.

- **Highlighted callouts** for important information

  Most participants and website users scan web pages to find the necessary information. Due to the lack of visual hierarchy, nearly every piece of content has the same level of importance. This makes it difficult to scan.

- **Enhanced search functionality** and visibility

During testing, only one participant utilized the search bar to find the information they needed. The search bar rests in the upper left corner of the screen, away from most of the content. By placing the search bar in a more central area and suggesting search queries, more users can complete tasks in less time and with more accuracy.